

Question #1 of 40

Question ID: 1474040

Which of the following statements is *least accurate*?

- A) The Sharpe ratio of a portfolio is unaffected by addition of cash or leverage in the portfolio. 
- B) Investors can take active risk that is suitable for them by investing in a combination of actively managed portfolio and benchmark portfolio. 
- C) A closet index fund has a low Sharpe ratio. 

Explanation

A closet index fund will have Sharpe ratio close to the benchmark's Sharpe ratio. The Sharpe ratio for a portfolio is indeed unaffected by addition of cash or leverage to the portfolio. However, information ratio does change as we add cash or leverage to the actively managed portfolio. Investors can combine benchmark portfolio and active portfolio to obtain optimal level of active risk for them.

(Module 40.2, LOS 40.b)

Question #2 of 40

Question ID: 1474057

Alisa Darent is evaluating several active portfolio managers with the same style and benchmark portfolio.

Manager	Active return	Active risk
Alfred	3.00%	12.00%
Brad	2.20%	11.00%
Charles	2.00%	10.50%

Benchmark return is expected to be 11%. What will be the maximum expected return for Darent's portfolio assuming that she wants to limit her active risk to 11%?

- A) 2.20% 
- B) 2.75% 
- C) 13.75% 

Explanation

Darent will select the manager with the highest information ratio – or Alfred.

$$IR(Alfred) = 3/12 = 0.25$$

$$IR(Brad) = 2.2/11 = 0.20$$

$$IR(Charles) = 2.0/10.50 = 0.19$$

$$\text{Expected active return} = E(R_A) = IR \times \sigma_A = 0.25 \times 11 = 2.75\%$$

$$\text{Expected return} = E(R_B) + E(R_A) = 11\% + 2.75\% = 13.75\%$$

(Module 40.3, LOS 40.d)

Question #3 of 40

Question ID: 1474066

Charles Griffith makes quarterly bets between stocks of industrial and utility sectors. The historical correlation between the returns of the two sectors is -0.20. Further information is as below:

Sector	E (R)	σ	Benchmark
			Weight
Industrial	12.00%	13.0%	80%
Utility	5.2%	2.5%	20%

The annualized active risk of Griffith's strategy is *closest* to:

A) 13.72%



B) 27.44%



C) 10.90%



Explanation

$$\text{Combined active risk} = \sigma_C = [\sigma_I^2 - 2\sigma_I\sigma_U r_{IU} + \sigma_U^2]^{1/2}$$

$$= [0.13^2 + 0.025^2 - 2(0.13)(0.025)(-0.20)]^{1/2} = 0.1372 \text{ or } 13.72\%$$

$$\text{Annualized active risk} = 0.1372 \times (4)^{1/2} = 0.2744 \text{ or } 27.44\%$$

(Module 40.4, LOS 40.e)

Question #4 of 40

Which of the following terms is the number of independent bets per year made by an active manager?

A) Information Coefficient



B) Transfer Coefficient



C) Breadth



Explanation

Breadth is the number of independent bets (based on unique information) made per year by the active manager.

(Module 40.3, LOS 40.c)

Question #5 of 40

Question ID: 1474055

An active manager currently covers 40 stocks and makes a forecast for each of them every quarter. Next year he intends to cover the same stocks but only once every 6 months. Assuming the manager's skill, measured in terms of the correlation of each forecast with actual returns doesn't change, which of the following statements is *most* accurate?

A) The information coefficient will fall by approximately 50%



B) The information ratio will fall by approximately 30%



C) The information ratio will fall by approximately 50%



Explanation

Information ratio (IR) = $IC \times \sqrt{BR}$

Hence a reduction in the breadth from 160 (40×4) to 80 (40×2) will cause an approximate 30% drop in the IR

With quarterly predictions $IR = IC \times 160^{1/2}$ = 12.65 (IC)

With semi-annual forecasts $IR = IC \times 80^{1/2}$ = 8.94 (IC)

$8.94IC / 12.65IC = 0.701$

Hence the Information Ratio will fall by approximately 30%. Note that full calculation is not required. Given that IR changes with the square root of breadth, a 50% drop in breadth must cause a less than 50% drop in IR. Note that it does not matter if the portfolio is constrained or unconstrained.

(Module 40.3, LOS 40.c)

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Question ID: 1474047

Which of the following terms is the ex-ante risk weighted correlation between forecasted active returns and actual active returns?

- A) Transfer Coefficient
- B) Information Coefficient
- C) Breadth



Explanation

Information coefficient is the ex-ante correlation between forecasted active returns and actual active returns. It captures the skill of the manager.

(Module 40.3, LOS 40.c)

Question #7 of 40

Question ID: 1474039

Which of the following statements is *least accurate*?

- A) Sharpe ratio of a portfolio consisting of a combination of benchmark and actively managed portfolio with positive active return will be higher than the Sharpe ratio of the benchmark.
- B) The information ratio of a constrained active portfolio is unaffected by aggressiveness of the active weights.
- C) Unlike Sharpe ratio, information ratio is affected due to addition of cash or leverage.



Explanation

Information ratio of an *unconstrained* active portfolio is unaffected by aggressiveness of the active weights. Sharpe ratio is unaffected by addition of cash or leverage but information ratio would be. A portfolio consisting of a combination of benchmark and an actively managed portfolio is calculated as:

$$SR_P^2 = SR_B^2 + IR^2$$

If the active return is positive, $IR > 0$ and $SR_P > SR_B$.

(Module 40.2, LOS 40.b)

Question #8 of 40

Question ID: 1474046

Which of the following terms is the cross-sectional correlation between forecasted active returns and actual active weights adjusted for risk?

A) Breadth



B) Transfer Coefficient



C) Information Coefficient



Explanation

Transfer coefficient = $TC = \text{CORR}(\mu_i/\sigma_i, \Delta w_i\sigma_i)$

(Module 40.3, LOS 40.c)

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Question ID: 1474065

Which of the following statements regarding the information ratio of an unconstrained portfolio is *most likely* correct?

A) A market timer who uses independent information to make predictions about market movements on a monthly basis and has an information ratio of 0.20 must have an information coefficient equal to a stock selector with the same information ratio who follows 10 stocks and revises his forecast quarterly



B) A market timer who uses independent information to make predictions about market movements on a monthly basis and has an information ratio of 0.20 must have an information coefficient lower than a stock selector with the same information ratio who follows 10 stocks and revises his forecast quarterly



C) A market timer who uses independent information to make predictions about market movements on a monthly basis and has an information ratio of 0.20 must have an information coefficient higher than a stock selector with the same information ratio who follows 10 stocks and revises his forecast quarterly



Explanation

Unconstrained Information ratio (IR) = $IC \times \sqrt{BR}$

Market timer: $0.20 = IC \times 12^{\frac{1}{2}}$ $IC = 0.20 / 3.464$ $IC = 0.058$

Selector: $0.20 = IC \times 40^{\frac{1}{2}}$ $IC = 0.20 / 6.325$ $IC = 0.032$

The market timer has a lower breadth. In order to achieve the same information ratio he must have a higher information coefficient. Note calculations not required.

(Module 40.4, LOS 40.e)

Question #10 of 40

Question ID: 1474038

Zeta fund has active return and active risk of 1.6% and 8% respectively. Benchmark portfolio has a Sharpe ratio of 0.35 and standard deviation of benchmark returns is 10.5%.

What is the level of active risk that an investor would need to take to maximize the Sharpe ratio of a portfolio consisting of Zeta fund and the benchmark portfolio?

A) 8%



B) 7%



C) 6%



Explanation

Information ratio (IR) = $1.6\% / 8\% = 0.2$

Optimal level of active risk = $\sigma^*_A = \frac{IR}{SR_B} \sigma_B = \frac{0.2}{0.35} (10.5) = 6\%$

Active risk of Zeta fund = 8%

Weight of Zeta fund = $6\% / 8\% = 0.75$

Weight of benchmark = 0.25

(Module 40.2, LOS 40.b)

Radina Radichkova, CFA, is considering investing in one of three actively managed funds whose benchmark is the FTSE 100. The Sharpe ratio and standard deviation of the benchmark are 0.50 and 15%, respectively.

Alpha Bankso Crystal

Active return 3.2% 2.8% 12.5%

Active risk 4.1% 3.6% 15.5%

Radichkova is also analyzing an actively managed portfolio consisting of financials and pharmaceuticals. The following table shows the weights and returns of the benchmark and portfolio:

Sector	WB	WP	RB	RP
Financials	50%	70%	9.2%	17.8%
Pharmaceuticals	50%	30%	8.1%	6.3%

Radichkova makes the following comments about the two-sector portfolio:

Comment 1: The value added is 5.7%.

Comment 2: The asset allocation decision only accounts for 0.22% of the value added.

Radichkova is writing a summary of the fundamental law of active management. In that section of the report, she states that the transfer coefficient can be viewed as the correlation between:

1. ex-ante active returns and actual active weights
2. risk-weighted optimal active weights and risk-weighted actual active weights
3. ex-ante active returns and realized active returns

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Question ID: 1474059

In relation to funds Alpha, Bankso, and Crystal, the highest achievable Sharpe ratio is *closest* to:

- A)** 0.85.
- B)** 0.90.
- C)** 0.95.



Explanation

First, find out which fund has the highest information ratio (IR) as $IR = \text{active return} / \text{active risk}$. This would be Crystal with an IR of 0.806. Then apply the following formula to discover the combined Sharpe ratio (SR):

$$SR_P^2 = SR_B^2 + IR^2$$

$$SR_P^2 = 0.52 + 0.8062 = 0.8996, SR_P = 0.9485$$

(Module 40.2, LOS 40.b)

Question #12 - 14 of 40

Question ID: 1474060

If the FTSE 100 and Crystal fund are combined in an optimal portfolio, what proportion should be invested in Crystal?

A) 136%.



B) 146%.



C) 156%.



Explanation

We first compute the optimal level of risk:

$$\sigma_A = \frac{IR}{SR_B} \sigma_B, \text{ therefore, } \sigma_A = \frac{0.806}{0.5} 0.15 = 0.242$$

This implies a weight in Crystal of $0.242 / 0.155 = 156\%$.

(Module 40.2, LOS 40.b)

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Question ID: 1474061

How many of Radichkova's comments are correct in relation to the two-sector portfolio?

A) One.



B) Both.



C) None.



Explanation

Both comments are correct.

The return of the portfolio is $70\% \times 17.8 + 30\% \times 6.3 = 14.35\%$.

The return of the benchmark is $50\% \times 9.2 + 50\% \times 8.1 = 8.65\%$.

Value added = $14.35 - 8.65 = 5.7\%$.

Return from asset allocation is $(70\% - 50\%) \times 9.2\% + (30\% - 50\%) \times 8.1\% = 0.22\%$ (i.e., active weights times benchmark returns).

This implies that return from stock selection is $5.7 - 0.22 = 5.48\%$.

Let us check that:

$70\% \times (17.8\% - 9.2\%) + 30\% \times (6.3\% - 8.1\%) = 5.48\%$ (i.e., portfolio weights times active returns).

(Module 40.1, LOS 40.a)

Question #14 - 14 of 40

Question ID: 1474062

Which are the correct definitions of the transfer coefficient included in Radichkova's report?

A) 1 and 2.



B) 2 and 3.



C) 1 and 3.



Explanation

The first two are correct definitions whereas number 3 is the definition of the information coefficient.

(Module 40.3, LOS 40.c)

Question #15 of 40

Question ID: 1474074

An active manager makes quarterly bets on the stocks in the Russell 2000 index and uses the index as the benchmark. The manager claims a modest IC of 0.02 using a stock screening model. Sam Fox, CFA makes the following two statements:

- I. The bets on the 2000 stocks in the index is not independent as the screens by definition introduce dependency in the decision process.
- II. The quarterly bets are likely to be independent.

How many of Fox's statements are correct:

- A) Only one statement is correct.
- B) Both statements are correct.
- C) Neither statement is correct.



Explanation

Fox is correct that screens (e.g., minimum dividend yield) would pass stocks with similar attributes and hence would introduce dependency in the decision making process. Fox is incorrect that the decisions over time are independent. Those stocks that pass the screen in one quarter are probably more likely to pass the same screen in the next quarter and hence the decisions are not truly independent.

(Module 40.4, LOS 40.f)

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Question ID: 1474033

Susan Thomas is evaluating the holdings of Primus fund. Based on the information below, the estimated active return is *closest* to:

Security (i)	Portfolio Weight (w_{Pi})	Benchmark Weight (w_{Bi})	ReturnE (R_i)
X	30%	40%	11.20%
y	15%	25%	4.25%
z	55%	35%	14.00%
Total	100%	100%	

- A) 1.77%



- B) 1.26%



C) 0.44%



Explanation

Portfolio return = $R_P = \sum(w_{Pi}) \times E(R_i) = 11.70\%$

Benchmark return = $R_B = \sum(w_{Bi}) \times E(R_i) = 10.44\%$

Active return = $R_P - R_B = 11.70\% - 10.44\% = 1.26\%$

(Module 40.1, LOS 40.a)

Question #17 of 40

Question ID: 1474048

Which of the following is correct for an unconstrained active portfolio?

A) $TC=1$



B) $TC>1$



C) $TC<1$



Explanation

$TC=1$ if the active portfolio has no constraints.

(Module 40.3, LOS 40.c)

Question #18 of 40

Question ID: 1508686

Helen Wilde is trying to estimate the active return of Optimal fund. A comparison of Optimal's holdings and that of the benchmark are shown below:

Asset Class (i)	Optimal Weight (w_{Pi})	Benchmark Weight (w_{Bi})	Optimal Return $E(R_{Pi})$	Benchmark Return $E(R_{Bi})$
Industrials	30%	40%	11%	12%
Financials	50%	30%	6%	5%
Utilities	20%	30%	14%	12%

The expected active return due to asset allocation for Optimal is *closest* to:

A) – 0.86%.



B) -0.44%.



C) – 1.40%.



Explanation

Asset Class (i)	Portfolio Weight (w_{Pi})	Benchmark Weight (w_{Bi})	Benchmark Return $E(R_{Bi})$	Active Return Weight (Δw_i)	(Δw_i) ($E(R_{Bi})$)
Industrials	30%	40%	12%	-10%	-1.20%
Financials	50%	30%	5%	20%	1.00%
Utilities	20%	30%	12%	-10%	-1.20%

(Module 40.1, LOS 40.a)

Charles Griffith makes quarterly bets between stocks of industrial and utility sectors. The historical correlation between the returns of the two sectors is -0.20 and Griffith's bets have been correct 55% of the time. Further information is as below:

Sector	E (R)	σ	Benchmark
			Weight
Industrial	12.00%	13.0%	80%
Utility	5.2%	2.5%	20%

The expected annualized active return of Griffith's sector rotation strategy is *closest* to:

- A) 10.64% 
- B) 13.72% 
- C) 5.48% 

Explanation

$$IC = 2(0.55) - 1 = 0.10$$

$$\text{Combined active risk} = \sigma_C = [\sigma_I^2 - 2\sigma_I\sigma_U r_{IU} + \sigma_U^2]^{1/2}$$

$$= [0.13^2 + 0.025^2 - 2(0.13)(0.025)(-0.20)]^{1/2} = 0.1372 \text{ or } 13.72\%$$

$$\text{Annualized active risk} = 0.1372 \times (4)^{1/2} = 0.2744 \text{ or } 27.44\%$$

$$\text{Annualized active return} = IC \times \sqrt{BR} \times \sigma_A = 0.10 \times (4)^{1/2} \times 0.2744 = 0.0548 \text{ or } 5.48\%$$

Alternatively,

Active return from this strategy using a probability weighted average (given Griffith makes correct calls 55% of time) of combined risk is:

$$(0.55)(0.1372) + (0.45)(-0.1372) = 0.0137 \text{ or } 1.37\% \text{ per quarter.}$$

$$\text{Annual active return} = 1.37\% \times 4 = 5.48\%.$$

(Module 40.4, LOS 40.e)

Question #20 of 40

Question ID: 1474053

An active manager has an information coefficient of 0.07, transfer coefficient of 0.90, and makes 49 independent bets per year. Benchmark portfolio has a Sharpe ratio of 0.40 and standard deviation of benchmark returns is 12%. The optimal amount of active risk is *closest* to:

A) 6%



B) 8%



C) 14%



Explanation

$$IR = (TC) IC \sqrt{BR} = (0.90) (0.07) \sqrt{49} = 0.441$$

For a constrained portfolio, the optimal level of residual risk can be computed as:

$$\sigma_A^* = (IR / SB_B) \sigma_B = (0.441 / 0.40)(0.12) = 13.23\%$$

(Module 40.3, LOS 40.c)

Question #21 of 40

Question ID: 1474063

Charles Griffith makes quarterly bets between stocks of industrial and utility sectors. Griffith's strategy has an annualized active risk of 18%. Based on the information below, If Griffith wants to limit his active risk to 6%, what is the allocation to Utility sector when Griffith is bullish about Industrial stocks?

Benchmark	
Sector	Weight
Industrial	80%
Utility	20%

A) 5%



B) -13%



C) 14%



Explanation

If active risk is limited to 6%, the deviation from the benchmark weights of 80% and 20% is limited to 6%/18% or 33%. Hence when Griffith is bullish about industrials, the weight to that sector will be 80% + 33% or 113% and the weight to utility sector will be 20% - 33% or -13%.

(Module 40.4, LOS 40.e)

Question #22 of 40

Question ID: 1474037

Zeta fund has active return and active risk of 1.6% and 8% respectively. Benchmark portfolio has a Sharpe ratio of 0.35 and standard deviation of benchmark returns is 10.5%.

The maximum possible Sharpe ratio of a portfolio consisting of Zeta fund and the benchmark portfolio is *closest* to:

A) 0.5



B) 0.4



C) 0.55



Explanation

$$SR_P = [SR_B^2 + IR^2]^{1/2} = [0.35^2 + 0.20^2]^{1/2} = 0.4031$$

(Module 40.2, LOS 40.b)

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Question ID: 1474064

Jon Gamlin is comparing a market timing strategy with a stock selection strategy. He draws the following two conclusions for unconstrained active managers:

Conclusion 1

To achieve the same information ratio, a market timer making weekly forecasts on the movement of the market needs to have a higher skill level than a stock selector following 25 stocks and updating the forecast semi-annually

Conclusion 2

A specialist following only 4 stocks who revises his forecast 100 times per year will achieve the same information ratio as a stock selector with the same skill level who follows 50 stocks and updates his assessments semi-annually

Regarding Gamlin's conclusions:

A) Only conclusion 2 is correct.



B) Neither conclusion is correct.



C) Only conclusion 1 is correct.



Explanation

In conclusion 1, the market timer has a breadth of 52 and the stock selector 50. In order to achieve the same information ratio, the stock selector would need to make up for the lower breadth with a higher information coefficient.

In conclusion 2, the specialist has a breadth of 400 and the selector 100. If they have the same skill level, the specialist with the larger breadth will have a higher information ratio

(Module 40.4, LOS 40.e)

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Question ID: 1474054

An active manager expects his information coefficient to drop from 0.08 to 0.02 in the coming period due to extremely volatile and unpredictable markets. As a response he intends to increase his breadth by a factor of 4. Which of the following statements is *most accurately* describes the impact on the information ratio?

A) The information ratio will remain constant



B) The information ratio will decrease



C) The information ratio will increase



Explanation

Information ratio (IR) = $IC \times \sqrt{BR}$

If breadth is increased by a factor of 4, this would increase the information ratio by a factor of 2. As the information coefficient is decreasing by a factor of 4, the information ratio will decrease.

(Module 40.3, LOS 40.c)

Question #25 of 40

Question ID: 1474036

Zeta fund has active return and active risk of 1.6% and 8% respectively. Benchmark portfolio has a Sharpe ratio of 0.35 and standard deviation of benchmark returns is 10.5%.

What is the weight of benchmark portfolio in a portfolio consisting of Zeta fund and the benchmark portfolio assuming that the portfolio is constructed to have optimal active risk?

A) 0.1667



B) 0.2



C) 0.25



Explanation

Information Ratio = active return / active risk = 1.6% / 8% = 0.2

$$\text{Optimal level of active risk} = \sigma_A^* = \frac{\text{IR}}{\text{SR}_B} \sigma_B = \frac{0.2}{0.35} (10.5) = 6\%$$

Active risk of Zeta fund = 8%

Weight of Zeta fund = 6% / 8% = 0.75

Weight of benchmark = 0.25

(Module 40.2, LOS 40.b)

Ufton Wealth Management's Ranger fund has proved popular with clients. An extract from the prospectus of the Ranger fund is shown in Exhibit 1.

Exhibit 1: Ranger Fund

Asset	Portfolio weight	Benchmark weight	Expected portfolio return	Expected benchmark return
U.S. equities	15%	20%	11%	9%
U.S. corporate bonds	35%	35%	8%	7%
International equities	8%	40%	14%	10%
U.S. real estate	42%	5%	7%	7%

Ufton awards its best performing fund manager with a large cash bonus each year. Details of the performance of three funds is shown in Exhibit 2. Risk-free rate is 2%.

Exhibit 2: Selected Fund Performance

Fund	Portfolio return	Benchmark return	Portfolio standard deviation	Benchmark standard deviation	Sharpe ratio	Tracking error
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Saltire	8.46%	5.80%	6.13%	4.50%	1.05	1.58%
Dragon	13.01%	11.56%	7.64%	5.15%	1.44	2.12%
Rose	11.39%	11.37%	11.01%	11.14%	0.85	0.21%

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Question ID: 1474042

The expected level of active return expected to be achieved by the Ranger fund is *closest* to:

- A) -9.00% 
- B) -0.09% 
- C) +3.49% 

Explanation

The expected active return achieved by a portfolio are calculated as the difference between the expected return on the portfolio and the expected return on the benchmark:

$$E(R_A) = \sum w_{Pi} \times E(R_{Pi}) - \sum w_{Bi} \times E(R_{Bi}) = E(R_P) - E(R_B)$$

Exhibit 1: Ranger Fund

Asset	Portfolio weight	Benchmark weight	Expected portfolio return	Expected benchmark return
U.S. equities	15%	20%	11%	9%
U.S. corporate bonds	35%	35%	8%	7%
International equities	8%	40%	14%	10%
U.S. real estate	42%	5%	7%	7%

$$E(R_P) = (0.15 \times 11\%) + (0.35 \times 8\%) + (0.08 \times 14\%) + (0.42 \times 7\%) = 8.51\%$$

$$E(R_B) = (0.20 \times 9\%) + (0.35 \times 7\%) + (0.40 \times 10\%) + (0.05 \times 7\%) = 8.60\%$$

$$E(R_A) = -0.09\%$$

(Module 40.1, LOS 40.a)

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Question ID: 1474043

The largest positive contribution to the active return achieved by the Ranger fund is expected to come from:

A) security selection.



B) asset allocation.



C) Cannot tell from the information available.



Explanation

The active return on a portfolio can be deconstructed to assess how much of the active return comes from active weighting, and how much comes from security selection.

The active return from active weighting is calculated by taking the sum of the active weights of each asset class multiplied by the benchmark return of the asset class:

$$R_{A(\text{from active weights})} = \sum \Delta w_i R_{Bi}$$

For the Ranger fund:

$$R_{A(\text{from active weights})} = (-0.05 \times 9\%) + (0 \times 7\%) + (-0.32 \times 10\%) + (0.37 \times 7\%) = -1.06\%$$

The active return from security selection is calculated by taking the sum of the weight each asset class in the portfolio multiplied by the difference in portfolio return on the asset class and the benchmark return on the asset class:

$$R_{A(\text{from security selection})} = \sum w_i (R_{Pi} - R_{Bi})$$

For the Ranger fund:

$$R_{A(\text{from security selection})} = (0.15 \times 2\%) + (0.35 \times 1\%) + (0.08 \times 4\%) + (0.42 \times 0\%) = 0.97\%$$

Of the total active return of -0.09%, active weighting has a negative contribution (1.06%), whereas security selection has a positive impact of 1.06%.

(Module 40.1, LOS 40.a)

Question #28 - 29 of 40

Question ID: 1474044

Of the three funds described in Exhibit 2, the fund with the highest information ratio is:

A) Saltire.



B) Dragon.



C) Rose.



Explanation

The information ratio measures the active return ($R_P - R_B$) per unit of active risk (tracking error). The information ratio for each fund is calculated as follows:

$$\text{Saltire: } (8.46\% - 5.80\%) / 1.58\% = 1.68$$

$$\text{Dragon: } (13.01 - 11.56\%) / 2.12\% = 0.68$$

$$\text{Rose: } (11.39\% - 11.37\%) / 0.21\% = 0.10$$

(Module 40.2, LOS 40.b)

Question #29 - 29 of 40

Question ID: 1474045

Of the three funds described in Exhibit 2, the *most* likely to be a closet index fund is:

A) Saltire.



B) Dragon.



C) Rose.



Explanation

A closet index fund is a fund, which is presented as being actively managed but covertly tracks the underlying benchmark index. It will achieve little active return and be exposed to little active risk, will have a low information ratio, and will have a Sharpe ratio close to the Sharpe ratio of the underlying benchmark. The Sharpe ratio is calculated as excess return over the risk-free asset per unit of portfolio risk: $(R_P - R_F) / \sigma_P$. Sharpe ratios for each fund's benchmark are calculated below $(R_B - R_F) / \sigma_B$:

Saltire: $(5.80\% - 2\%) / (4.50\%) = 0.84$

Dragon: $(11.56\% - 2\%) / (5.15\%) = 1.86$

Rose: $(11.37\% - 2\%) / (11.14\%) = 0.84$

The Rose fund has the lowest information ratio of the three funds, and its Sharpe ratio (0.85) is very close to that of its benchmark (0.84). It is therefore most likely to be a closet index fund.

(Module 40.2, LOS 40.b)

Question #30 of 40

Question ID: 1474052

Which of the following is correct for a constrained active portfolio?

A) $TC < 1$



B) $TC > 1$



C) $TC = 1$



Explanation

When we impose constraints on portfolios, the actual active weights (Δw_i) will differ from optimal active weights (Δw_i^*) and $TC < 1$.

(Module 40.3, LOS 40.c)

Sundar Mithai, CFA, is a fund manager for Pearl Investments and makes a monthly report to the firm's partners. Mithai mentions two active managers in his report, Galab and Phasar. Exhibit 1 provides additional information on the two managers:

Exhibit 1: Selected Information on Galab and Phasar

	Galab	Phasar
Information coefficient	0.22	0.37
Transfer coefficient	0.8	0.73
Active risk	5.6%	6.6%
Active return	10.8%	9.2%

Mithai makes the following comments regarding the two active managers:

Comment 1: The investment mandate of Phasar appears to be less constrained relative to Galab.

Comment 2: Galab appears to have better skill at predicting returns.

Mithai recently decided to give all the analysts at the firm a refresher on the fundamental law of active portfolio management. Details of a hypothetical unconstrained fund is shown in Exhibit 2.

Exhibit 2: Hypothetical Fund

Information coefficient	0.14
Monthly active bets	5
Active risk	4.32%

According to the fundamental law of active management, how many forecasts is Galab making per month?

A) 3.



B) 10.



C) 36.



Explanation

The extended law states that:

active return = $TC \times IC \times \sqrt{BR} \times \text{active risk}$

$10.8\% = 0.8 \times 0.22 \times \sqrt{BR} \times 5.6\%$

$BR = 120$ (annual), Galab is making 10 bets per month.

(Module 40.4, LOS 40.e)

Question #32 - 34 of 40

Question ID: 1474071

How many of Mithai's comments are correct in relation to the comparison between Galab and Phasar?

A) One.



B) Both.



C) None.



Explanation

Both comments are incorrect. Phasar has higher information coefficient, which indicates better skill at predicting results. Phasar has also a lower transfer coefficient, which indicates that it is a more restrained fund.

(Module 40.3, LOS 40.c)

Question #33 - 34 of 40

Question ID: 1474072

The expected active return generated by the hypothetical fund described in Exhibit 2 is:

A) 3.12%.



B) 4.68%.



C) 8.20%.



Explanation

The expected level of active return achieved by a portfolio is calculated as follows:

$$E(R_A) = TC(IC) \sqrt{BR} \sigma_A$$

where:

TC = transfer co-efficient

IC = information co-efficient

BR = number of independent active bets taken per year

σ_A = active risk

In an unconstrained portfolio, the transfer co-efficient is equal to 1. Therefore the active return generated by the fund will be:

$$E(R_A) = 1 \times 0.14 \times \sqrt{60} \times 4.32\% = 4.68\%$$

(Module 40.3, LOS 40.c)

Question #34 - 34 of 40

Question ID: 1493686

If the hypothetical fund described in Exhibit 2 was subject to investment constraints, its expected active return would be expected to:

A) rise.



B) fall.



C) remain unchanged.



Explanation

If an actively managed portfolio is not subject to investment constraints, its transfer co-efficient will be equal to 1, reflecting the manager's ability to achieve optimal active weight in the portfolio. If constraints are imposed, the transfer co-efficient will be between 0 and 1. Given active return is positively related to the transfer co-efficient, the imposition of constraints must lead to a reduction in expected active return.

(Module 40.3, LOS 40.c)

Question #35 of 40

Question ID: 1474051

An active manager has an information coefficient of 0.05 and makes 36 independent bets per year. What is the manager's information ratio given a transfer coefficient of 0.75?

- A) 0.23
- B) 0.45
- C) 1.35

**Explanation**

Information ratio =

$$IR = (TC) IC\sqrt{BR} = (0.75) (0.05) \sqrt{36} = 0.225$$

(Module 40.3, LOS 40.c)

Question #36 of 40

Question ID: 1474067

Tom Grenkin is a market timer with an information ratio of 0.75. He makes a prediction of the movement in the market each quarter. Jane Fortina is a stock selector who follows 50 companies and revises her assessment each quarter. She also has an information ratio of 0.75. Assuming both managers have unconstrained portfolios, which of the following statements regarding the two managers is *most accurate*?

- A) As both managers have the same information ratio, they must also have the same information coefficient.
- B) As Fortina's strategy has a much larger breadth, she must have a larger information coefficient than Grenkin.
- C) As Grenkin makes fewer bets per year, he requires a higher information coefficient on each bet than Fortina to achieve the same information ratio.

**Explanation**

$$(IR) = IC \times \sqrt{BR}$$

As a stock selector, Fortina makes many more bets per period and has a much larger breadth. She therefore requires a lower information coefficient than Grenkin to achieve the same information ratio. Grenkin requires a higher coefficient.

$$\text{Grenkin } 0.75 = IC \times 4^{\frac{1}{2}} \quad IC = 0.75/2 \quad = 0.375$$

$$\text{Fontina } 0.75 = IC \times 200^{\frac{1}{2}} \quad IC = 0.75/14.14 \quad = 0.053$$

(Note: Calculations are not required)

(Module 40.4, LOS 40.e)

Question #37 of 40

Question ID: 1474056

When choosing an active manager, an investor with a high level of risk aversion:

- A) will choose the manager with the highest active return. 
- B) will choose a manager with the lowest active risk. 
- C) will choose the manager with the highest information ratio. 

Explanation

Value added is independent of the level of risk aversion. All investors will choose the manager with the highest information ratio. Those with higher levels of risk aversion will implement the strategy less aggressively (i.e., invest a larger proportion in the benchmark portfolio).

(Module 40.3, LOS 40.d)

Question #38 of 40

Question ID: 1474050

An active manager has an information coefficient of 0.08, transfer coefficient of 0.50, and makes 100 independent bets per year. What is the expected active return for an active risk constraint of 5%?

- A) 2.4% 
- B) 2.0% 
- C) 1.8% 

Explanation

$$E(R_A) = (TC) IC \sqrt{BR} \sigma_A = (0.50) (0.08) \sqrt{100} (0.05) = 0.02 \text{ or } 2\%$$

(Module 40.3, LOS 40.c)

Question #39 of 40

Question ID: 1474035

Helen Wilde is trying to estimate the active return of Optimal fund. A comparison of Optimal's holdings and that of the benchmark are shown below:

Asset Class (i)	Optimal Weight (w_{Pi})	Benchmark Weight (w_{Bi})	Optimal Return $E(R_{Pi})$	Benchmark Return $E(R_{Bi})$
Industrials	30%	40%	11%	12%
Financials	50%	30%	6%	5%
Utilities	20%	30%	14%	12%

The expected active return for Optimal is *closest* to:

A) -0.44%



B) - 1.40%



C) - 0.80%



Explanation

Portfolio return = $R_P = \sum(w_{Pi}) \times E(R_{Pi}) = 9.10\%$

Benchmark return = $R_B = \sum(w_{Bi}) \times E(R_{Bi}) = 9.90\%$

Active return = $R_P - R_B = 9.10\% - 9.90\% = -0.80\%$

(Module 40.1, LOS 40.a)

Question #40 of 40

Question ID: 1474075

Pamela Grieve claims that her information coefficient is 0.20 on monthly bets on 10 stocks in the healthcare industry. Assuming unconstrained optimization, the reduction in her information ratio if her bets have a correlation coefficient of 0.45 as opposed to being truly independent is *closest* to:

A) 22%



B) 45%



C) 86%



Explanation

Grieve's breadth assuming independent bets = $10 \times 12 = 120$

Information ratio assuming independent bets = $IC\sqrt{BR} = 0.20 \times \sqrt{120} = 2.19$

If the bets are correlated, $BR =$

$$\frac{N}{1+(N-1)r} = \frac{120}{1+(120-1)0.45} = 2.20$$

New information ratio assuming correlated bets $IC\sqrt{BR} = 0.20 \times \sqrt{2.20} = 0.30$

% reduction = $1 - 0.30/2.19 = 86.4\%$

(Module 40.4, LOS 40.f)